- *Vertical scalability* —Scaling vertically, or scaling up or down, means adding hardware resources to or removing them from the computing node, such as CPU or memory. This approach is limited, since it's not possible to keep adding hardware resources. On the other hand, applications don't need to be explicitly designed to be scaled up or down.
- *Horizontal scalability* —Scaling horizontally, or scaling out or in, means adding more computing nodes or containers to, or removing them from, the system. This approach doesn't have the same limits as vertical scalability, but it requires applications to be scalable.

Traditional systems would usually adopt vertical scalability in the event of increasing workloads. Adding CPU and memory was a common approach to enable an application to support more users without redesigning it for scalability. This is still a good option in specific scenarios, but we need something else for the cloud.

In the cloud, where everything is dynamic and in constant change, horizontal scalability is preferred. Thanks to the abstraction levels offered by the cloud computing models, it's straightforward to spin up new instances of your application rather than increasing the computational power of the machines already running. Since the cloud is elastic, we can scale application instances in and out quickly and dynamically. I discussed elasticity as one of the main characteristics of the cloud: computing resources can be provisioned and released proactively, depending on the demand. Scalability is a prerequisite for elasticity.

Figure 1.5 shows the difference between vertical and horizontal scalability. In the first case, we scale by adding more resources to the existing virtual